

Notes on the $\mathfrak{h}_3$ -Generated Estimation Algebra Tower and Cubic Sensor Model

Daniyal Hameedi

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Introduction

This note develops the algebraic machinery needed to classify exact and quasi-finite filtering solutions for polynomial sensor models, with the cubic sensor as the target case where Ocone's theorem rules out an exact finite-dimensional filter. §1 recovers the Kalman filter from representation-theoretic first principles as a proof of concept for the method, and establishes — via the anti-automorphism ω of $U(\mathfrak{h}_3)$, defined intrinsically rather than through a choice of inner product — a rigorous version of the Gram-matrix/null-vector argument that will be reused throughout. §2–4 build the corresponding machinery for the cubic sensor: the K_m^j generator algebra, its triangular decomposition, and a corrected general commutator formula. §5 uses this machinery to computationally construct the Lie subalgebra generated by $\{L_0, x^3\}$ to depth 32, and finds — and, in §5.7, proves rather than merely observes — a quadratic growth law, a period-5 zero-weight structure, and an exact $\Delta \mapsto -1 - \Delta$ reflection symmetry governed by ω . These results characterize the bracket-generated subalgebra itself in detail, but do not yet constitute a filterability classification: that requires relating this growth data to the Gelfand–Kirillov dimension of $D(1)/\text{Ann}(\rho)$, which is the invariant Bernstein's inequality actually constrains. Closing that gap (Open Question 1 below) is the primary outstanding obstacle in the programme.

1 Preliminary Result: The Kalman Filter

In this section we present a proof-of-concept result whereby the Kalman Filter equations are obtained from representation theoretic concerns of the associated algebra. The Linear Gaussian Signal/Observation model is given by:

$$\frac{dY}{dt} = Cx + R^{1/2} \frac{dV}{dt} \quad (1)$$

$$\frac{dx}{dt} = Ax + Q^{1/2} \frac{dW}{dt} \quad (2)$$

Where W, V are random walks, x is the state-variable, Y is the signal. It possesses the following estimation algebra $\mathcal{E}_{K.B.}$:

$$\mathcal{E}_{K.B.} = \{\mathbb{I}, x, \partial_x, \mathcal{L}_0\}, \quad \mathcal{L}_0 = \frac{1}{2} \partial_x^2 - Cx \partial_x + C - \frac{1}{2} A^2 x^2 \quad (3)$$

We begin with $\mathfrak{h}_3$:

$$[x, \partial_x] = \mathbb{I}, \quad [x, x] = [\partial_x, \partial_x] = 0, \quad (4)$$

and consider that the estimation algebra is completely spanned at $F_2^B(\mathcal{U}(\mathfrak{h}_3)) \implies k = 2$. We consider then the Filtrations:

$$F_1^B(\mathcal{U}(\mathfrak{h}_3)) = \{\mathbb{I}, x, \partial_x\} \quad (5)$$

We can then construct the Gram matrix at $j' = 1$:

$$G^{(1)} = \begin{pmatrix} \langle u | \mathcal{B}_0^\dagger \mathcal{B}_0 | u \rangle & \langle u | \mathcal{B}_0^\dagger \mathcal{B}_1 | u \rangle & \langle u | \mathcal{B}_0^\dagger \mathcal{B}_2 | u \rangle \\ \langle u | \mathcal{B}_1^\dagger \mathcal{B}_0 | u \rangle & \langle u | \mathcal{B}_1^\dagger \mathcal{B}_1 | u \rangle & \langle u | \mathcal{B}_1^\dagger \mathcal{B}_2 | u \rangle \\ \langle u | \mathcal{B}_2^\dagger \mathcal{B}_0 | u \rangle & \langle u | \mathcal{B}_2^\dagger \mathcal{B}_1 | u \rangle & \langle u | \mathcal{B}_2^\dagger \mathcal{B}_2 | u \rangle \end{pmatrix} \quad (6)$$

We additionally consider a generic Annihilator condition:

$$\begin{aligned} (\alpha \mathcal{B}_0 + \beta \mathcal{B}_1 + \mathcal{B}_2) \cdot u &= 0 \\ \implies (\alpha \mathcal{B}_0 + \beta \mathcal{B}_1 + \mathcal{B}_2) &= 0 \\ \implies \mathcal{B}_2 &= (-\alpha \mathcal{B}_0 - \beta \mathcal{B}_1) \end{aligned} \quad (7)$$

Which is an operator identity of the state u . This then allows for $\det(G^{(1)})$ to vanish:

$$\begin{aligned} \det(G^{(1)}) &= \langle \bar{\mathcal{B}}_0 \mathcal{B}_0 \rangle (\langle \bar{\mathcal{B}}_1 \mathcal{B}_1 \rangle \langle \bar{\mathcal{B}}_2 \mathcal{B}_2 \rangle - \langle \bar{\mathcal{B}}_1 \mathcal{B}_2 \rangle \langle \bar{\mathcal{B}}_2 \mathcal{B}_1 \rangle) \\ &\quad - \langle \bar{\mathcal{B}}_0 \mathcal{B}_1 \rangle (\langle \bar{\mathcal{B}}_1 \mathcal{B}_0 \rangle \langle \bar{\mathcal{B}}_2 \mathcal{B}_2 \rangle - \langle \bar{\mathcal{B}}_1 \mathcal{B}_2 \rangle \langle \bar{\mathcal{B}}_2 \mathcal{B}_0 \rangle) \\ &\quad + \langle \bar{\mathcal{B}}_0 \mathcal{B}_2 \rangle (\langle \bar{\mathcal{B}}_1 \mathcal{B}_0 \rangle \langle \bar{\mathcal{B}}_2 \mathcal{B}_1 \rangle - \langle \bar{\mathcal{B}}_1 \mathcal{B}_1 \rangle \langle \bar{\mathcal{B}}_2 \mathcal{B}_0 \rangle) \end{aligned} \quad (8)$$

though we first take a detour, elucidating the nature of the adjoint $\bar{\mathcal{B}}_m$:

1.1 The anti-automorphism ω and the adjoint annihilator condition

Rather than positing an inner product on a representation of $U(\mathfrak{h}_3)$ — which would implicitly select a class of representations for the Gram matrix construction — we define the dagger operation via the principal anti-automorphism ω of $U(\mathfrak{h}_3)$ itself, acting purely at the level of the algebra.

Let $\omega : U(\mathfrak{h}_3) \rightarrow U(\mathfrak{h}_3)$ be the linear map fixed on generators by

$$\omega(X) = X, \quad \omega(P) = -P, \quad \omega(I) = I, \quad (9)$$

and extended to all of $U(\mathfrak{h}_3)$ by anti-multiplicativity, $\omega(ab) = \omega(b)\omega(a)$, together with linearity. We write $B_n^\dagger := \omega(B_n)$.

Proposition 1. ω is a well-defined algebra anti-automorphism of $U(\mathfrak{h}_3)$ satisfying $\omega^2 = \text{id}$.

Proof. It suffices to check consistency with the defining relation (3). We compute

$$\omega([X, P]) = \omega(XP - PX) = \omega(P)\omega(X) - \omega(X)\omega(P) = (-P)X - X(-P) = XP - PX = [X, P], \quad (10)$$

so that $\omega(I) = I$ is *forced* by anti-multiplicativity together with (3), rather than imposed independently. Since ω is an involution on generators extended anti-multiplicatively, $\omega^2 = \text{id}$ follows immediately. $\square$

Crucially, ω reverses order but does *not* annihilate the Heisenberg commutator: for $B_1 = X$, $B_2 = P$,

$$[\omega(B_1), B_2] = [X, P] = I \neq 0, \quad (11)$$

so no additional commutativity assumption between barred and unbarred generators is introduced. In particular, for the concrete realization $B_1 = x$, $B_2 = \partial_x$ used in Section 3,

$$\bar{B}_0 = B_0 = I, \quad \bar{B}_1 = \omega(B_1) = B_1 = x, \quad \bar{B}_2 = \omega(B_2) = -B_2 = -\partial_x. \quad (12)$$

The sign assignment $\omega(P) = -P$ (rather than the equally consistent alternative $\omega(X) = -X$, $\omega(P) = P$) is fixed by requiring ω to reproduce the formal adjoint of (x, ∂_x) under the flat bilinear pairing $\langle f, g \rangle = \int f(x)g(x) dx$, for which integration by parts gives ∂_x formal anti-symmetry and multiplication by x formal symmetry. This selects the class of unitarily-equivalent (Stone–von Neumann) Schrödinger-type representations; extension of ω to indecomposable, non-semisimple representations of $\mathfrak{h}_3 \rtimes \mathfrak{sp}(2)$ is left for future work.

Since ω is linear, applying it to the annihilator condition (16),

$$B_2 = -(\alpha B_1 + \beta B_0), \quad (13)$$

gives, using $\bar{B}_2 = -B_2$, $\bar{B}_1 = B_1$, $\bar{B}_0 = B_0$, the *adjoint* annihilator relation

$$\bar{B}_2 = \alpha B_1 + \beta B_0, \quad (14)$$

with the *same* coefficients α, β — no independent bra-side coefficients need be introduced. With this machinery established we can now show that $\det(G^{(1)}) = 0$ rigorously with the Null Vector constraint.

1.2 Vanishing of $\det G^{(1)}$

Let $N_0 := \langle u|u \rangle$, $p := \langle u|B_1|u \rangle$, $m_2 := \langle u|B_1^2|u \rangle$. Substituting the ket-side annihilator condition (16) and the adjoint condition (14) directly — with no further assumptions — every remaining matrix element reduces to N_0, p, m_2 :

$$\langle \bar{B}_2 B_2 \rangle = \alpha^2 m_2 + 2\alpha\beta p + \beta^2 N_0, \quad (15)$$

$$\langle \bar{B}_1 B_2 \rangle = \langle \bar{B}_2 B_1 \rangle = -(\alpha m_2 + \beta p) =: K. \quad (16)$$

The three cofactor terms of (8) (cf. Eq. 17) become

$$A = \langle \bar{B}_0 B_0 \rangle (\langle \bar{B}_1 B_1 \rangle \langle \bar{B}_2 B_2 \rangle - K^2) = \beta^2 N_0 (m_2 N_0 - p^2), \quad (17)$$

$$B = -\langle \bar{B}_0 B_1 \rangle (\langle \bar{B}_1 B_0 \rangle \langle \bar{B}_2 B_2 \rangle - K \langle \bar{B}_2 B_0 \rangle) = \alpha\beta p (m_2 N_0 - p^2), \quad (18)$$

$$C = \langle \bar{B}_0 B_2 \rangle (\langle \bar{B}_1 B_0 \rangle \langle \bar{B}_2 B_1 \rangle - \langle \bar{B}_1 B_1 \rangle \langle \bar{B}_2 B_0 \rangle) = -\beta(\alpha p + \beta N_0)(m_2 N_0 - p^2). \quad (19)$$

Summing,

$$\det G^{(1)} = A + B + C = (m_2 N_0 - p^2) [\beta^2 N_0 + \alpha\beta p - \alpha\beta p - \beta^2 N_0] = 0. \quad (20)$$

The vanishing is a genuine consequence of the annihilator condition (16), not an identity holding for arbitrary u : repeating the computation *without* substituting (16) or (14) — treating $\langle B_2 \rangle, \langle B_2^2 \rangle, \langle B_1 B_2 \rangle$ as independent moments q, m'_2, c — yields

$$\det G^{(1)} \Big|_{\text{generic}} = -m_2 m'_2 + c^2 + c + p^2 m'_2 - 2pcq - pq + m_2 q^2 \neq 0, \quad (21)$$

confirming that $\det G^{(1)} = 0$ correctly detects the existence of the level-1 null vector rather than holding vacuously.

Therefore, $j' = 1$, meaning that $\dim(\mathcal{O}_k) = 3$ according to the formula ??, and we can compute the following Wei-Norman Decomposition[1]:

$$u = e^{\alpha x^2 + \beta x + \gamma} \quad (22)$$

Indeed, we can now project out into the appropriate tangent space for all elements in $F_2^B(\mathcal{U}(\mathfrak{h}_3))$:

$$\frac{\partial u}{\partial a} = x^2 u, \quad \frac{\partial u}{\partial b} = x u, \quad \frac{\partial u}{\partial c} = u \quad (23)$$

It is trivial to verify that the tangent space relation is true for the identity $\mathbb{I}$. We can now begin the more mechanical aspect of the proof:

Proof.

$$\begin{aligned} x_i u &= \frac{\partial u}{\partial b}, \\ \partial_i u &= (2ax + b)u = \left(a \frac{\partial u}{\partial b} + b \frac{\partial u}{\partial c} \right) \\ x^i x^j u &= \frac{\partial u}{\partial a_{ij}} \\ \partial_i \partial_j u &= \partial_i (2a_{ji} x^i + b_j) u = (2a_{ji} + 4a^2 x^2 - 4abx + b^2) u = (2a + b^2 + 4a^2 \frac{\partial u}{\partial a} - 4ab \frac{\partial u}{\partial b}) u \\ x_i \partial_j u &= x_i (ax + b) u = (ax^2 + bx) u = \left(a \frac{\partial u}{\partial a} + b \frac{\partial u}{\partial b} \right) u \end{aligned}$$

Under suitable assumptions of completeness, we take it that this extends to the full group. $\square$

1.3 Filter Equations

We consider that all filter equations are governed by the Zakai/DMZ equation:

$$\partial_t u(x, t) = \mathcal{H}(t) \cdot u(x, t) \quad (24)$$

for $u(x, t) = u(x; a(t), b(t), c(t))$ for $u \in \mathcal{G}/\text{Stab}(H_3 \rtimes Sp(2))$. The RHS of the Zakai Equation is:

$$\partial_t u = \frac{\partial u}{\partial a} \left(\dot{a} + \frac{1}{2} \frac{C^2}{R} \right) + \frac{\partial u}{\partial b} \dot{b} + \frac{\partial u}{\partial c} \dot{c} \quad (25)$$

Which is a projection into the tangent space, and where $\dot{a}$ has picked up an Ito correction. To compute $\mathcal{H}(t)$ we must consider the non-compact form of the dynamics generator:

$$\mathcal{H}(t) = \mathcal{L}^* + x \frac{C}{R} \frac{dY}{dt} \quad (26)$$

We recall that for Linear Gaussian models:

$$\frac{dY}{dt} = Cx + R^{1/2} \frac{dV}{dt} \quad (27)$$

$$\frac{dx}{dt} = Ax + Q^{1/2} \frac{dW}{dt} \quad (28)$$

Where W, V are random walks.

$$\begin{aligned}
\mathcal{H}(t)u &= \mathcal{L}^*u + x \frac{C}{R} \left(\frac{dY}{dt} \right) u \\
&\Rightarrow \mathcal{H}(t)u = -\partial_x(Axu) + \frac{1}{2}Q\partial_x^2u + x \frac{C}{R} \left(\frac{dY}{dt} \right) u \\
&\Rightarrow \mathcal{H}(t)u = -Au - Ax\partial_xu + \frac{1}{2}Q(2a + b^2 + 4a^2\frac{\partial u}{\partial a} + 4ab\frac{\partial u}{\partial b})u + x \frac{C}{R} \left(\frac{dY}{dt} \right) u \\
&\Rightarrow \mathcal{H}(t)u = -Au - A \left(2a\frac{\partial u}{\partial a} + b\frac{\partial u}{\partial b} \right) u + \frac{1}{2}Q(2a + b^2 + 4a^2\frac{\partial u}{\partial a} + 4ab\frac{\partial u}{\partial b})u + \frac{\partial u}{\partial b} \left(\frac{C}{R} \frac{dY}{dt} \right) u
\end{aligned}$$

We now collect terms:

$$\mathcal{H}(t) = \left(-2Aa + \frac{1}{2}Q4a^2 \right) \partial_a u + \left(-Ab + \frac{1}{2}Q4ab + \frac{C}{R} \frac{dY}{dt} \right) \partial_b u + \left(-A + \frac{1}{2}2a + b^2 \right) \partial_c u \quad (29)$$

We can then match coefficients:

$$\frac{\partial u}{\partial a} \dot{a} = \left(-2Aa + 2Qa^2 - \frac{1}{2} \frac{C^2}{R} \right) \partial_a u \quad (30)$$

$$\frac{\partial u}{\partial b} \dot{b} = \left(-Ab + 2Qab + \frac{C}{R} \frac{dY}{dt} \right) \partial_b u \quad (31)$$

$$\frac{\partial u}{\partial c} \dot{c} = (-A + a + b^2) \partial_c u \quad (32)$$

We now perform the coordinate transformation, taking the generic polynomial coefficients to functions of the covariance P and filter mean m :

$$a = \frac{-1}{2P}, b = \frac{m}{P} \implies \partial_a u = \frac{\partial u}{\partial P} \frac{\partial P}{\partial a} = 2P^2 \frac{\partial u}{\partial P}, \partial_b u = \frac{\partial u}{\partial m} \frac{\partial m}{\partial b} + \frac{\partial u}{\partial P} \frac{\partial P}{\partial b} = P \frac{\partial u}{\partial m} + \frac{m}{P} \frac{\partial u}{\partial P} \quad (33)$$

Which leads us to the Riccati equation directly from the equation in $\dot{a}$:

$$\dot{P} = 2AP + Q - \frac{C^2}{R} P^2 \quad (34)$$

The second equation becomes:

$$\begin{aligned}
\left(\frac{1}{P} \dot{m} - \frac{m}{P^2} \dot{P} \right) &= \left(-\frac{Am}{P} - \frac{Qm}{P^2} \right) + \frac{C}{R} \frac{dY}{dt} \\
\text{Substituting for } \dot{P}: \dot{m} &= \left(-A - \frac{C^2}{R} P \right) m dt + \frac{PC}{R} dY \\
\implies dm &= Amdt + \frac{PC}{R} (dY - Cmdt) \\
&\implies \dot{m} = Am + \frac{PC}{R} \left(\frac{dY}{dt} - Cm \right) \quad (35)
\end{aligned}$$

which we can see is simply the filter mean equation. The final equation for $\dot{c}$ is simply a normalization tracking equation, but can undergo a similar treatment.

2 The Cubic Sensor

The cubic sensor model is famously associated with an infinite dimensional estimation algebra, and we state it as follows [2]:

$$\frac{dY}{dt} = x^3 + \frac{dV}{dt} \quad (36)$$

$$\frac{dx}{dt} = \frac{dW}{dt} \quad (37)$$

Where V, W are Random Walks, and $h(x) = x^3$ is manifestly non-linear.

The Estimation Algebra is then initially spanned by:

$$\mathcal{E}_{C.S.} = \{\mathbb{I}, x^3, \mathcal{L}_0\}, \quad \mathcal{L}_0 = \frac{1}{2}\partial_x^2 \quad (38)$$

Taking the commutator action between the two non-identity elements is as follows:

$$[\mathcal{L}_0, x^3] = \frac{1}{2}[\partial_x^2, x^3] = 3x + 3x^2\partial_x. \quad (39)$$

We $\mathcal{E}_{C.S.}$ to sit within an ambient object, which we now specify.

Consider the Heisenberg Algebra $\mathfrak{h}_3$:

$$[x, \partial_x] = \mathbb{I}, \quad [x, x] = 0, \quad [\partial_x, \partial_x] = 0 \quad (40)$$

We then consider the Universal Enveloping Algebra in a suitable PBW basis of the Heisenberg Algebra: $x^\alpha P^\beta \mathbb{I}^\gamma \in \mathcal{U}(\mathfrak{h}_3)$ equipped with the Bernstein Filtration $F_j^B(\mathcal{U}(\mathfrak{h}_3))$:

$$F_j^B(\mathcal{U}(\mathfrak{h}_3)) = \{x^\alpha \partial_x^\beta \mathbb{I}^\gamma \in \mathcal{U}(\mathfrak{h}_3), \quad s.t. \alpha + \beta + \gamma = j\}. \quad (41)$$

This sort of graded algebra possesses a known convergence in its Hilbert series:

$$H(t) = \frac{1}{(1-t)^2} \quad (42)$$

3 The Universal Enveloping Algebra of $\mathfrak{h}_3$

We now construct an expedient symbolic language for examining the representation theory of the universal enveloping algebra of $\mathfrak{h}_3$. Generically, the Universal Enveloping algebra is infinite, and by Ocone's theorem, we know that for the cubic and higher polynomial sensor models, there is no finite dimensional representation for the Zakai Equation. However, in this section we aim to establish whether there exist finite recurrences within the Universal Enveloping Algebra, and whether this implies Quasi-finiteness of the representation.

We consider the generalized action for $K_m^{(j)} = x^{m-j}\partial_x^j$ leads to the following algebra:

$$[K_m^j, K_n^i] = \left(\sum_{k=1}^j \binom{j}{k} \frac{(n-i)!}{(n-i-k)!} K_{m+n-2k}^{i+j-k} \right) - \left(\sum_{l=1}^i \binom{i}{l} \frac{(m-j)!}{(m-j-l)!} K_{m+n-2l}^{i+j-l} \right) \quad (43)$$

We consider that this has the following Cartan sub-algebra $\mathfrak{h}$:

$$\mathfrak{h} = \{\mathbb{I}, K_{2a}^a = x^a \partial_x^a\} \quad (44)$$

For the particular choice $K_2^1 = x\partial_x$ we can see that it is a grading ('Cartan') operator and we select only this one for our purposes:

$$[x\partial_x, K_m^j] = (m - 2j)K_m^j \quad (45)$$

This allows us to make the following association for sectors of the space of generators, leading to a Triangular decomposition:

$$(m - 2j) < 0, K_m^j \in \mathfrak{g}_- \quad (46)$$

$$K_0^0, K_{2a}^a \in \mathfrak{g}_0 \quad (47)$$

$$(m - 2j) > 0, K_m^j \in \mathfrak{g}_+ \quad (48)$$

Where the identity is a trivial zero of the grading. We select K_0^0 and K_2^1 specifically to be in $\mathfrak{g}_0$ as all higher order operators form a non-semi-simple zero-weight space, and we make the choice for K_2^1 to be the true grading element.

If we assert the highest weight state condition:

$$K_m^j |\Lambda\rangle = 0, K_m^j \in \mathfrak{g}_+ \quad (49)$$

We then see that generically, the associated Verma module $\mathcal{V} = U(\mathfrak{g}_-) |\Lambda\rangle$ is infinite dimensional, due to each weight space in $m - 2j < 0$ having infinite multiplicity in m, j . We choose to re-order this by imposing a maximal complexity filtration for each K_m^j in it's Bernstein degree m (a natural choice given that the generators are composed of differential operators), and consider that the pair $(m, m - 2j)$ now parametrizes the generator space, where $m - 2j$ indicates the weight-coordinate, and m indicates the total complexity bound for a particular weight-space.

Notably, imposing the filtration does not lose algebraic data, but merely re-orders the problem tractably, and ensures that our filtered Verma Module $F^B\mathcal{V} = F^BU(\mathfrak{g}_-) |\Lambda\rangle$ is now finite dimensional at a given weighting and complexity.

We identify some further useful and distinguished sub-algebras of this family. For elements of this algebra, we then identify the action with a generic K_m^j via commutator:

$$[K_n^1, K_m^j] = (m - jn)K_{m+n-2}^j - \sum_{l=2}^j \binom{j}{l} \frac{(n-1)!}{(n-1-l)!} K_{n+m-2l}^{j+1-l} \quad (50)$$

Which is a finite depth recurrence relation. We note that the leading term here has preserved Bernstein degree. We denote this as our horizontal transport operator T_n which moves elements within a given degree of the filtration. Indeed we can see that a Witt Sub-algebra is obeyed by the transport operators:

$$[T_n, T_m] = (m - n)T_{m+n-2} \quad (51)$$

We then examine $i = 2$:

$$\begin{aligned} [K_n^2, K_m^j] &= (2m - nj)K_{m+n-2}^{j+1} + \left((m-j)(m-j-1) - \frac{j(j-1)}{2}(n-2)(n-3) \right) K_{m+n-4}^j \\ &\quad + \sum_{l=3}^j \binom{j}{l} (n-2)_l K_{m+n-2l}^{2+j-l} \end{aligned} \quad (52)$$

Which we can see is a spin growth operator by inspection of the leading term which grows the Bernstein degree $j \rightarrow j + 1$. We now denote such operators as Γ_n , and that these operators satisfy the following algebra with each other:

$$[\Gamma_n, \Gamma_m] = 2(m - n)K_{m+n-2}^3 + ((m-3) - 3(n-2)(n-3))T_{m+n-4} \quad (53)$$

Additionally we note that there is an exceptional locus at $2m - nj = 0$, which we will interpret as a natural surface in the module at which we expect null vectors, truncation, or closure to appear.

3.1 Finite Recursions

In this subsection, we now use the distinguished +1 spin shift operator family Γ_n to try and formulate finite recurrences under the appropriate adjoint action.

We construct our generic reference state for a fixed weight Δ as:

$$K_2^1 |\phi\rangle = \Delta |\phi\rangle \quad (54)$$

Indeed, we can now consider the fixed weight sector such that:

$$\text{ad}_{K_2^1} K_m^j |\phi\rangle = [K_2^1, K_m^j] |\phi\rangle = (m - 2j) |\phi\rangle = r |\phi\rangle \quad (55)$$

Where $\text{ad}_{K_2^1} K_m^j = [K_2^1, K_m^j]$ is the adjoint action via the grading generator. We can rearrange such that $m = r + 2j$ expresses the Bernstein complexity of each generator in terms of the grading r and spin j . We define the following candidate descendant states:

$$v_j = K_{r+2j}^j |\phi\rangle \quad (56)$$

Note that we take $r < 0$ as we wish to have positive modes annihilate: $U(\mathfrak{g}_+) |\phi\rangle = 0$ as a condition for quasi-finiteness. We wish to find a finite recursion by examining the action of $\text{ad}_{K_n^2} = \Gamma_n(\cdot)$ on these descendants:

$$\begin{aligned} \Gamma_n(v_j) = [\Gamma_n, K_{r+2j}^j] |\phi\rangle &= (2r + (4 - n)j) K_{r+2j+n-2}^{j+1} |\phi\rangle \\ &+ \left((r + j - 1) - \frac{j(j-1)}{2} (n-2)(n-3) \right) K_{r+2j+n-4}^j |\phi\rangle \\ &+ \sum_{l=3}^j \binom{j}{l} (n-2)_l K_{r+2j+n-2l}^{2+j-l} |\phi\rangle \end{aligned} \quad (57)$$

Indeed, we consider the relation for $v_{j+i+1} = \Gamma_n(v_{j+i})$ only for the leading order term:

$$v_{i+j+1} = [\Gamma_n, K_{r^{(i)}+2j}^{j+i}] = \left[\prod_{l=0}^i (2r + l(n-4) + (4-n)j) \right] K_{r+(i+1)(n-2)+2j}^{j+i+1} \quad (58)$$

By considering $c_i = \prod_{l=0}^i (2r + l(n-4) + (4-n)j) = 0$, we realize that we can formulate an appropriate lead term vanishing (resonance) condition at a particular value ℓ^* :

$$\ell^* = j - \frac{2r}{n-4}, \quad \ell^* \in \mathbb{Z}^+ \quad (59)$$

This also imposes a maximal spin at $J_{max} = j + \ell^*$ after which point every higher leading order spin state vanishes. We now consider that a similar analysis of the sub-leading term will lead to a second resonance, whereby we expect the coefficient

$$A(m, j, n) = \left((m-j)(m-j-1) - \frac{j(j-1)}{2} (n-2)(n-3) \right)$$

to vanish. Indeed we are led to a second constraint by examining only the sub-leading order term:

$$v_{j+i+1} = \Gamma_n(v_{j+i}) = \prod_{\ell=0}^i \left((m + \ell(n-4) - j)(m + \ell(n-4) - j - 1) - \frac{j(j-1)}{2}(n-2)(n-3) \right) K_{m+(i+1)(n-4)}^j \quad (60)$$

We now consider for some special ℓ' the required condition for the leading coefficient to vanish:

$$\ell' = \frac{1}{2} \left(\frac{(1 - 2(m-j))}{(n-4)} \pm \left[\frac{\sqrt{1 + 2j(j-1)(n-2)(n-3)}}{(n-4)} \right] \right) \quad (61)$$

We can see that ℓ' is divergent for $n = 4$, with special loci identified at $n = 2, 3$, $j = 0, 1$, $m = j$. By now imposing that $\ell' < \ell^* = \ell' + \ell_c$, $\ell_c \in \mathbb{Z}^+$, we consider that the following expression for ℓ_c can find candidate representation weights:

$$\ell_c = \frac{2j(n-3) - 2r - 1 \mp \sqrt{S}}{2(n-4)}, \text{ where } S = 1 + 2j(j-1)(n-2)(n-3) \quad (62)$$

We can rearrange to:

$$r = j(n-3) - \frac{1}{2} - \ell_c(n-4) \mp \frac{1}{2} \sqrt{1 + 2j(j-1)(n-2)(n-3)} \quad (63)$$

Which tells us the representation weights where the sub-leading term will vanish ℓ_c adjoint actions before the leading term. We consider that there exist several immediately obvious singular point values of $n = 2, 3, 4$ which we can choose:

$$(n=4), \quad r = j - \frac{1}{2} \mp \frac{1}{2} \sqrt{1 + 4j(j-1)} \quad (64)$$

$$(n=3), \quad r = -\frac{1}{2} + \ell_c \mp \frac{1}{2} \quad (65)$$

$$(n=2), \quad r = -j - \frac{1}{2} + 2\ell_c \mp \frac{1}{2} \quad (66)$$

We consider now that families of representation are then described by the integer triple (n, ℓ_c, j) which then uniquely defines the corresponding weight family r . We note that these resonance conditions are results about fixed filtration slices as opposed to the whole infinite algebra. We note that at $n = 4$ the weight loses dependence on the stagger term ℓ_c .

We then consider that Γ_4 is a zero-weight operator:

$$[T_2, \Gamma_4] = 0\Gamma_4 \implies r = 0. \quad (67)$$

Which acts on generic K_m^j :

$$[\Gamma_4, K_m^j] = (2m - 4j)K_{m+2}^{j+1} + (m^2 - m(2j+1) + 2j)K_m^j \quad (68)$$

Where the tail vanishes for $j < 3$.

4 General Commutator formula

Working in the Bernstein-filtered $U(\mathfrak{h}_3)$ with generators $K_m^j = x^{m-j} \partial_x^j$, direct computation corrects the sub-leading coefficient appearing in the Γ_n -family commutators. The general result is

$$[K_n^2, K_m^j] = (2m - nj) K_{n+m-2}^{j+1} + [(m-j)(m-j-1) - \frac{1}{2}j(j-1)(n-2)(n-3)] K_{n+m-4}^j - \sum_{l=3}^j \binom{j}{l} (n-2)_l K_{n+m-2l}^{2+j-l}, \quad (69)$$

where $(n-2)_l = (n-2)(n-3)\cdots(n-1-l)$ is the falling factorial. This differs from the naive extrapolation of the leading-term recursion by the replacement $j(j-1)$ (not $j(j+1)$) in the sub-leading coefficient, and by the full $(m-j)(m-j-1)$ quadratic factor (not the bare linear term $m-j-1$). The correction was verified independently against direct computation of $[x^a \partial_x^b, x^c \partial_x^d]$ on three separate test cases spanning different (n, j, m) .

4.1 Specialization to $n = 2$: the physical generator

Since $(n-2) = 0$, both the sub-leading $(n-2)(n-3)$ term and the entire tail sum vanish identically, leaving the exact two-term recursion for $\Gamma_2 = 2L_0$:

$$[\Gamma_2, K_m^j] = (2m - 2j) K_{m-2}^{j+1} + (m-j)(m-j-1) K_{m-2}^j. \quad (70)$$

This reproduces the estimation algebra's defining bracket exactly: at $j = 0, m = 3$, (70) gives $[L_0, x^3] = 3K_1^0 + 3K_3^1 = 3x + 3x^2 \partial_x$, matching direct computation.

4.2 Specialization to $n = 4$: the Cartan-family case

At $n = 4$, $(n-2)(n-3) = 2$, giving

$$[\Gamma_4, K_m^j] = (2m - 4j) K_{m+2}^{j+1} + [(m-j)(m-j-1) - j(j-1)] K_m^j, \quad (71)$$

for $j < 3$ (tail vanishing). Crucially, $\Gamma_4 = K_4^2 = x^2 \partial_x^2$ is itself a member of the Cartan family $\{K_{2a}^a\}$, hence has weight zero under the grading operator K_2^1 . Every bracket $[\Gamma_4, K_m^j]$ therefore *automatically* preserves the weight $r = m - 2j$ of its argument — a fact following directly from weight-additivity under $\text{ad}_{K_2^1}$, independent of the explicit form of (71). This is the structural origin of the divergence at $n = 4$ in the original drift-counting resonance formulas (the parameter ℓ , counting iterated weight-shifting steps, is ill-defined for a weight-preserving operator); the correct resonance condition at $n = 4$ is instead the direct vanishing of the sub-leading eigenvalue, $(m-j)(m-j-1) - j(j-1) = 0$, with no iteration count required.

5 The ECS-generated tower

5.1 Construction

Let $\text{ECS} = \{I, x^3, L_0\}$ with $L_0 = \frac{1}{2} \partial_x^2 = \frac{1}{2} K_2^2$. Representing every element as a finite formal sum $\sum c_{j,m} K_m^j$, the two generators act on any such element via the closed recursions (70) (for L_0) and

$$[x^3, K_m^j] = - \sum_{k=1}^{\min(j,3)} \binom{j}{k} (3)_k K_{m+3-2k}^{j-k} \quad (72)$$

(for x^3 , via Leibniz expansion of $[x^3, \partial_x^j]$). Starting from $\{L_0, x^3\}$, the Lie subalgebra of $U(\mathfrak{h}_3)$ generated by iterated brackets was computed to depth 32, maintaining a genuine linear basis via exact (rational) Gaussian elimination at each depth rather than enumerating the exponentially many bracket words directly.

5.2 Dimension growth

An initial read of the depth-20 data (2, 3, 5, 7, 10, 12, 16, 19, 23, 27, 30, 33, 37, 40, ...) suggested roughly linear growth. Extending to depth 32 and applying a proper ratio test rules this out: $\dim(d)/d$ does not converge to a constant but climbs steadily (from 2.0 at $d = 1$ to 6.4 at $d = 32$, with no sign of flattening), whereas $\dim(d)/d^2$ converges cleanly to ≈ 0.20 by $d \approx 20$ and remains stable through $d = 32$. A direct polynomial fit confirms this quantitatively: a quadratic fits the cumulative dimension data with roughly $40\times$ lower residual sum of squares than a linear fit.

Proposition 2 (Empirical). *The cumulative dimension of the ECS-generated Lie subalgebra grows quadratically in bracket-depth, $\dim(d) \sim d^2/5$, not linearly.*

The coefficient $1/5$ is plausibly connected to the period-5 weight-conservation structure of Section 2.3 below, rather than being a coincidental fitting artifact: both the number of distinct weights reached by depth d (which grows as $O(d)$, from the $d + 1$ -slot argument of Section 2.3) and the multiplicity of independent generators found within a single weight (which also grows with d rather than staying bounded, per direct inspection of the per-weight generator counts) increase with depth; the product of two linearly-growing quantities is the natural mechanism for an overall quadratic law. This has not yet been proven directly from the bracket combinatorics and should be treated as a well-evidenced empirical regularity, not an established theorem.

This growth rate is a statement about the bracket-generated Lie subalgebra itself and should *not* be conflated with the Gelfand–Kirillov dimension of $D(1)/\text{Ann}(\rho)$ used elsewhere in this programme: the two are computed via different filtrations (bracket-depth of a Lie subalgebra versus the PBW/Bernstein degree filtration of an associative module), and for a rank-1 Weyl-algebra module the latter is constrained by Bernstein’s inequality to the strict dichotomy $\text{GKdim} \in \{1, 2\}$ (holonomic versus Ocneanu’s generic obstruction), with no intermediate values — so no numerical growth exponent found here can be read off directly as a Gelfand–Kirillov dimension.

5.3 Periodicity of the zero-weight (Cartan-family) hits

Observation 1. *The tower intersects the zero-weight space $\mathfrak{g}_0 = \text{span}\{I, K_2^1, K_4^2, K_6^3, \dots\}$ non-trivially only at depths that are multiples of 5, and does so at every such depth through the range checked (5, 10, 15, 20).*

This follows from weight conservation: $x^3 = K_3^0$ has weight $+3$, L_0 has weight -2 (bracketing lowers the weight of the state it acts on by 2). A bracket word using x^3 exactly a times and L_0 exactly b times has total weight $3a - 2b$, which vanishes iff $a = 2k$, $b = 3k$ for integer k — forcing word length (depth) $a + b = 5k$. Each successive zero-weight hit adds exactly one new Cartan-family generator to the reachable span: $\{I, K_2^1, K_4^2\}$ at depth 5, adding K_6^3 at depth 10, K_8^4 at depth 15, K_{10}^5 at depth 20.

5.4 Resonance on the highest-weight state

Acting on the formal highest-weight vector $|\phi\rangle = x^\Delta$ (with $K_2^1|\phi\rangle = \Delta|\phi\rangle$ and $K_{2a}^a|\phi\rangle = \Delta(\Delta - 1)\cdots(\Delta - a + 1)|\phi\rangle$), each zero-weight tower vector evaluates to an explicit polynomial in Δ . Scanning all zero-weight hits through depth 30:

Proposition 3 (Empirical). *Every zero-weight tower vector at depth $\equiv 0 \pmod{10}$ (depths 10, 20, 30 checked) admits $\Delta = -\frac{1}{2}$ as an exact rational root, alongside a complex-conjugate pair. No zero-weight tower vector at depth $\equiv 5 \pmod{10}$ (depths 5, 15, 25 checked) admits any real root.*

This is a specific, checkable resonance statement intrinsic to the ECS-generated subalgebra, distinct from the generic (n, ℓ_c, j) -parametrized resonance lattice obtained from the abstract Γ_n -descendant construction on a free highest-weight state.

Test of the identification $r = \Delta + \frac{1}{2}$. Under this shift, the confirmed resonance $\Delta = -\frac{1}{2}$ maps exactly to $r = 0$, matching the note's own $n = 3$, $\ell_c = 1$ result (its eq. 29). This match is exact and recurs without exception across all nine independent zero-weight vectors found at depths 10, 20, and 30. However, two caveats prevent this from being treated as a confirmed identification of the full lattice:

- (i) The same numerical value is also produced by the $n = 2$ family (its eq. 30, $r = (1 - j)/3$) at the specific spin $j = 1$, so the match alone does not distinguish which abstract family the tower is realizing.
- (ii) An explicit search of every root (rational and algebraic irrational) of every zero-weight polynomial through depth 30 against both the $n = 3$ ladder ($\ell_c = 2, 3, \dots$, predicting $\Delta = -\frac{1}{6}, \frac{1}{6}, \frac{1}{2}, \dots$) and the $n = 2$ ladder ($j = 0, 2, 3, \dots$, predicting $\Delta = -\frac{1}{6}, -\frac{5}{6}, -\frac{7}{6}, \dots$) found no matches beyond the single $\Delta = -\frac{1}{2}$ point, despite the search covering polynomials up to degree 6.

The honest state of the identification is therefore: exact and repeated confirmation at one point, ambiguous between two candidate families, and no evidence yet of the extended ladder structure either family would predict.

5.5 Γ_4 as a chain-extension operator

Applying Γ_4 (via (71)) directly to the tower's non-zero-weight homogeneous vectors does not return an eigenvector, but consistently extends the vector's support by one further step along the same $(j, m) \rightarrow (j + 1, m + 2)$ index pattern already present in the state (tested at depths 6, 9, 11, 12, 14). This is consistent with Γ_4 acting as a spin-raising “ladder” generator organizing each fixed-weight slice of the tower, analogous to a screening operator in the Coulomb-gas construction of degenerate Verma modules, though closure of Γ_4 on a fixed weight space into a self-contained algebra has not yet been separately established.

5.6 An exact reflection symmetry and the complex locus $\text{Re}(\Delta) = -\frac{1}{2}$

Examining the roots of every zero-weight resonance polynomial (not merely whether $\Delta = -\frac{1}{2}$ is among them) reveals a sharp structural fact. Every root, at every depth checked (5, 10, 15, 20, 25, 30; 16 independent vectors in total, of degrees 2 through 7), has real part *exactly* $-\frac{1}{2}$ — not approximately, and not only for the previously-noted $\Delta = -\frac{1}{2}$ resonance itself, but for every complex root as well.

Proposition 4. *For every zero-weight tower vector checked, the resonance polynomial $P(\Delta)$ satisfies the exact identity*

$$P(\Delta) = (-1)^{\deg P} P(-1 - \Delta), \quad (73)$$

verified symbolically with zero residual in all 16 cases across depths 5–30.

Since $\Delta = -\frac{1}{2}$ is the unique fixed point of the involution $\Delta \mapsto -1 - \Delta$, a polynomial obeying (73) has its entire root set symmetric about $-\frac{1}{2}$: real roots land exactly on it, and complex roots occur in conjugate pairs straddling it — accounting for both the isolated real resonance at depths $\equiv 0 \pmod{10}$ and the purely complex root sets at depths $\equiv 5 \pmod{10}$ as two faces of the same symmetry, rather than as separate phenomena. Under the identification $r = \Delta + \frac{1}{2}$ proposed above, (73) is equivalent to the statement that r is always purely imaginary (or zero) across the entire zero-weight resonance structure.

This symmetry is proven only empirically at present (verified case-by-case, not derived from a general argument), but a natural candidate mechanism is already present in this programme: the anti-automorphism ω ($\omega(x) = x$, $\omega(\partial) = -\partial$) established as the correct adjoint structure in the companion D-module work. Reflection dualities of the form $\Delta \leftrightarrow -1 - \Delta$ are the expected signature of such a parity/adjoint map acting on a Bernstein–Sato-type functional equation; whether ω applied directly to a zero-weight tower vector reproduces the $\Delta \rightarrow -1 - \Delta$ -transformed polynomial exactly has not yet been checked, and would upgrade (73) from an exact but unexplained empirical symmetry to a direct consequence of already-established structure.

5.7 A proof of the reflection symmetry via ω

Recall from §1.1 the anti-automorphism ω of $U(\mathfrak{h}_3)$ fixed on generators by $\omega(x) = x$, $\omega(\partial_x) = -\partial_x$, extended by $\omega(ab) = \omega(b)\omega(a)$. We show that the reflection symmetry of Proposition 4 is not merely an empirical regularity but a direct consequence of ω , together with an explicit description of how ω acts on every layer of the tower — not only the zero-weight one.

Step 1: ω fixes both generators. For $K_m^j = x^{m-j} \partial_x^j$ with $a := m-j \geq 0$, anti-multiplicativity together with the standard normal-ordering identity

$$\partial_x^j x^a = \sum_{k=0}^{\min(a,j)} \binom{j}{k} (a)_k x^{a-k} \partial_x^{j-k}, \quad (a)_k := a(a-1) \cdots (a-k+1),$$

gives

$$\omega(K_m^j) = (-1)^j \partial_x^j x^a = (-1)^j \sum_{k=0}^{\min(a,j)} \binom{j}{k} (a)_k K_{m-2k}^{j-k}. \quad (74)$$

For $L_0 = K_2^2$ we have $a = 0$, so only $k = 0$ survives and $\omega(L_0) = L_0$. For $x^3 = K_3^0$ we have $j = 0$, so again only $k = 0$ survives and $\omega(x^3) = x^3$. Both generators of the ECS tower are exactly ω -fixed.

Step 2: ω reverses the sign of a bracket. Anti-multiplicativity gives, for any $A, B \in U(\mathfrak{h}_3)$,

$$\omega([A, B]) = \omega(AB) - \omega(BA) = \omega(B)\omega(A) - \omega(A)\omega(B) = -[\omega(A), \omega(B)]. \quad (75)$$

Proposition 5 (ω -grading of the tower). *Every basis vector produced at bracket-depth d of the ECS tower — i.e. every $W_d = [g_d, W_{d-1}]$ with $g_d \in \{L_0, x^3\}$, iterated from the depth-1 seeds, of any weight — satisfies*

$$\omega(W_d) = (-1)^{d-1} W_d$$

exactly, for every $d \geq 1$.

Proof. Induction on d . For $d = 1$, $W_1 \in \{L_0, x^3\}$ and Step 1 gives $\omega(W_1) = W_1 = (-1)^0 W_1$. Suppose $\omega(W_{d-1}) = (-1)^{d-2} W_{d-1}$. By (75) and Step 1,

$$\omega(W_d) = \omega([g_d, W_{d-1}]) = -[\omega(g_d), \omega(W_{d-1})] = -[g_d, (-1)^{d-2} W_{d-1}] = (-1)^{d-1} [g_d, W_{d-1}] = (-1)^{d-1} W_d.$$

This holds regardless of which generator is chosen as g_d at each step, so it holds for the entire depth- d layer, and hence for any linear combination of depth- d vectors — in particular for every zero-weight vector arising at depth d . $\square$

Step 3: the resonance polynomial reflects under ω . Restricting (74) to a zero-weight generator K_{2j}^j ($a = j$) gives

$$\omega(K_{2j}^j) = (-1)^j \sum_{k=0}^j \binom{j}{k} (j)_k K_{2j-2k}^{j-k}.$$

Evaluating on the highest-weight state $|\phi\rangle = x^\Delta$, where $K_{2j'}^{j'}$ acts as $(\Delta)_{j'}$, and applying the Vandermonde convolution identity for falling factorials, $(x+y)_n = \sum_{k=0}^n \binom{n}{k} (x)_k (y)_{n-k}$, with $x = y = j$, collapses the sum to a single closed form:

$$\omega(K_{2j}^j) \Big|_{x^\Delta} = (-1)^j (\Delta+1)(\Delta+2) \cdots (\Delta+j) = (\Delta)_j \Big|_{\Delta \rightarrow -1-\Delta}. \quad (76)$$

By linearity, for any zero-weight vector $v = \sum_j c_j K_{2j}^j$ with resonance polynomial $P(\Delta) = \sum_j c_j (\Delta)_j$,

$$\omega(v) \Big|_{x^\Delta} = P(-1-\Delta).$$

Theorem 1 (Proof of Proposition 4). *Let v be any zero-weight tower vector arising at bracket-depth d , with resonance polynomial $P(\Delta)$. Then*

$$P(-1-\Delta) = (-1)^{d-1} P(\Delta).$$

Proof. By the Proposition above, $\omega(v) = (-1)^{d-1} v$ as an element of $U(\mathfrak{h}_3)$. Evaluating both sides on x^Δ and applying (76) term-by-term to the left side gives $P(-1-\Delta) = (-1)^{d-1} P(\Delta)$. $\square$

Since $\deg P = d/5 + 1$ at each zero-weight depth $d = 5k$ (Observation 1: each period adds exactly one Cartan-family generator to the span), the depth-based eigenvalue $(-1)^{d-1}$ and the degree-based form $(-1)^{\deg P}$ stated in §5.6 agree automatically, since $d-1 \equiv \deg P \pmod{2}$ for every k . This also explains Proposition 3 in full: at $d \equiv 0 \pmod{10}$, $\deg P$ is odd, so P has a real root by the intermediate value theorem regardless of any symmetry — and the ω -symmetry then forces every real root to land exactly at $-\frac{1}{2}$, the unique fixed point of the involution $\Delta \mapsto -1-\Delta$. At $d \equiv 5 \pmod{10}$, $\deg P$ is even and no real root is forced, matching the purely-complex root sets observed there.

[Scope] The theorem above is stated for the tower exactly as constructed — basis vectors built as left-nested words $[g_d, [g_{d-1}, \dots]]$ using a fresh copy of L_0 or x^3 at each step, matching

the actual generation procedure in Appendix A. This is not automatically the same as the fully Lie-closed subalgebra (which would also bracket two already-composite elements W_i, W_j directly against each other); the eigenvalue statement above would need to be checked separately if full closure is taken.

5.8 Test of Γ_4 as a period-advancing operator (falsified)

Motivated by the depth-mod-5 periodicity, it is natural to ask whether the zero-weight span at depth $10(k+1)$ is generated from the span at depth $10k$ by repeated application of Γ_4 alone — i.e., whether Γ_4 furnishes a single finite generator connecting successive periods. This was tested directly: the full cumulative basis at depth 10 (27-dimensional) was bracketed once with Γ_4 , producing 23 new independent directions, and the combined span was checked against the actual depth-20 cumulative basis (79-dimensional).

Observation 2. *Of the 23 new directions produced by a single Γ_4 application to the depth-10 basis, only approximately 2 overlap with the genuinely new content of the depth-20 basis; the residual (79 minus what is explained) drops only from 52 (the no- Γ_4 baseline, $79 - 27$) to 50. A second iterated Γ_4 application does not improve this. The hypothesis that Γ_4 alone generates each successive period from the previous one is therefore false: Γ_4 's established role (Section 5.6) is a narrow, weight-preserving chain-extension, and it does not reach the bulk of the new weights that a fresh round of L_0/x^3 brackets introduces at each period.*

6 Open questions

1. **(Highest priority.)** Whether $\Delta = -\frac{1}{2}$ corresponds to an actual holonomic point, i.e. whether $\text{Ann}(x^{-1/2}) \neq 0$ as an ideal of $D(1)$ — not merely whether a specific tower-generated element has eigenvalue zero on the single state $x^{-1/2}$. This is the check that would convert the numerical resonance into a genuine $\text{GKdim} = 1$ classification point.
2. Whether Γ_4 (and/or K_2^1) is reachable from $\{L_0, x^3\}$ *individually*, rather than only in the fixed combinations found at each zero-weight hit. Two independent depth-5 combinations already allow elimination of the identity component (yielding $\Gamma_4 + 2K_2^1$ cleanly); a further independent relation at any depth ≥ 5 divisible by 5 would fully triangulate the pair. (Note: Γ_4 has separately been ruled out as a *period-advancing* generator taking one zero-weight block to the next — see Section 5.8 — but this narrower reachability question remains open.)
3. Whether $[\Gamma_4, \Gamma_4\text{-conjugates}]$ closes into a recognizable algebra (Heisenberg, $\mathfrak{sl}_2$, or similar) when restricted to a fixed weight space — the computation that would convert the chain-extension observation above into an explicit multiplicity/composition-length statement for $D(1)/\text{Ann}(\rho)$ at each resonant weight.
4. Structural (rather than curve-fitted) proof of the empirical $\dim(d) \sim d^2/5$ growth law, ideally by directly establishing that both the number of distinct weights reached and the multiplicity within each weight grow linearly in depth, and of the depth-mod-5 zero-weight periodicity and depth-mod-10 $\Delta = -\frac{1}{2}$ resonance, via the same weight-counting argument used to establish the period-5 law in the first place.
5. If a finite generating mechanism for successive zero-weight blocks exists at all, it is not Γ_4 alone (Section 5.8); identifying what (if anything) plays this role — plausibly requiring

a second, independent generating direction to account for the linearly-growing number of new weights per period — remains open.

6. Connection of any resolved resonance/classification structure back to the physical parameters of the cubic sensor (noise/SNR scaling entering Δ), which has not yet been addressed at any stage of this computation.

References

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A ECS tower generation: code and full results

A.1 Method

The generators are represented as finite formal sums $\sum c_{j,m} K_m^j$ (Python dictionaries keyed by (j, m) , with exact `Fraction` coefficients throughout — no floating point is used anywhere, so every zero reported is an exact algebraic zero). Starting from the seed pair $\{L_0, x^3\}$, at each depth every current basis vector is bracketed with L_0 (via the closed recursion (70)) and with x^3 (via its Leibniz-expansion recursion), and the resulting candidates are reduced against the accumulated basis by exact Gaussian elimination, keeping only the linearly independent directions. This avoids the 2^{depth} blow-up of naively enumerating bracket words, most of which turn out to be linearly dependent on each other.

A.2 Code

Listing 1: ECS tower generator with exact linear-algebra basis reduction.

```

1 from fractions import Fraction as F
2 from math import comb
3
4 def bracket_L0(v):
5     # [L0, K^j_m] = (m-j) K^{j+1}_m + (1/2)(m-j)(m-j-1) K^j_{m-2}
6     out = {}
7     for (j, m), c in v.items():
8         u = m - j
9         out[(j+1, m)] = out.get((j+1, m), F(0)) + c*u
10        out[(j, m-2)] = out.get((j, m-2), F(0)) + c*F(u*(u-1), 2)
11    return {k: c for k, c in out.items() if c != 0}
12
13 def falling(n, k):
14     r = 1
15     for i in range(k): r *= (n - i)
16     return r

```



```

17
18 def bracket_x3(v):
19     #  $[x^3, K^j_m] = -\sum_{k=1}^{\min(j,3)} C(j,k) (3)_k K^{j-k}_{m+3-2k}$ 
20     out = {}
21     for (j, m), c in v.items():
22         for k in range(1, min(j,3)+1):
23             coeff = -comb(j,k) * falling(3,k)
24             key = (j-k, m+3-2*k)
25             out[key] = out.get(key, F(0)) + c*coeff
26     return {k: c for k, c in out.items() if c != 0}
27
28 def reduce_basis(vectors, basis):
29     """Row-reduce new vectors against existing basis; return the
30     newly independent ones (exact Gaussian elimination over Q)."""
31     all_keys = set()
32     for v in basis + vectors:
33         all_keys.update(v.keys())
34     keys = sorted(all_keys)
35     def to_row(v): return [v.get(k, F(0)) for k in keys]
36
37     rows = [to_row(b) for b in basis]
38     new_indep = []
39     for v in vectors:
40         row = to_row(v)
41         r = row[:]
42         for br in rows:
43             piv = next((i for i, x in enumerate(br) if x != 0), None)
44             if piv is not None and r[piv] != 0:
45                 factor = F(r[piv], br[piv])
46                 r = [ri - factor*bi for ri, bi in zip(r, br)]
47         if any(x != 0 for x in r):
48             rows.append(r)
49             new_indep.append(v)
50     return new_indep
51
52 L0 = {(2,2): F(1,2)}
53 X3 = {(0,3): F(1)}
54
55 basis = [L0, X3]
56 frontier = [L0, X3]
57
58 MAX_DEPTH = 32
59 for depth in range(2, MAX_DEPTH+1):
60     candidates = []
61     for v in frontier:
62         vl, vx = bracket_L0(v), bracket_x3(v)
63         if vl: candidates.append(vl)
64         if vx: candidates.append(vx)
65     new_indep = reduce_basis(candidates, basis)
66     basis.extend(new_indep)
67     frontier = new_indep
68     print(f"depth {depth}: {len(new_indep)} new, "
69           f"cumulative dim = {len(basis)}")

```

Zero-weight (Cartan-family) hits are identified by checking, for each newly independent vector, whether every (j, m) key in its support satisfies $m = 2j$. Resonance on the highest-weight state $|\phi\rangle = x^\Delta$ is obtained by substituting $K_{2a}^a \mapsto \Delta(\Delta - 1) \cdots (\Delta - a + 1)$ into each zero-weight vector and solving the resulting polynomial in Δ symbolically (sympy).

A.3 Full results, depths 1–32

Table 1: Dimension growth of the ECS-generated Lie subalgebra of $U(\mathfrak{h}_3)$, and the depths at which it intersects the zero-weight (Cartan-family) space.

Depth	New independent gen.	Cumulative dim.	Note
1	2	2	seed (L_0, x^3)
2	1	3	–
3	2	5	–
4	2	7	–
5	3	10	zero-weight hit (Δ : complex roots only)
6	2	12	–
7	4	16	–
8	3	19	–
9	4	23	–
10	4	27	zero-weight hit ($\Delta = -\frac{1}{2}$ exact root)
11	3	30	–
12	3	33	–
13	4	37	–
14	3	40	–
15	5	45	zero-weight hit (Δ : complex roots only)
16	6	51	–
17	6	57	–
18	7	64	–
19	8	72	–
20	7	79	zero-weight hit ($\Delta = -\frac{1}{2}$ exact root)
21	7	86	–
22	8	94	–
23	7	101	–
24	8	109	–
25	10	119	zero-weight hit (Δ : complex roots only)
26	10	129	–
27	11	140	–
28	13	153	–
29	12	165	–
30	13	178	zero-weight hit ($\Delta = -\frac{1}{2}$ exact root)
31	13	191	–
32	13	204	–

The zero-weight hits occur at exactly the depths predicted by the weight-conservation argument of Section 2.3 ($5 \mid \text{depth}$, with no exceptions), and the $\Delta = -\frac{1}{2}$ resonance occurs at exactly the depths $10 \mid \text{depth}$ among these, again with no exceptions across the three periods checked (10, 20, 30) and no false positives at the intervening period-5 depths (5, 15, 25).